

Task 2 – Intermediate Level

Python Programs with Sample Outputs

1. Web Scraper: Extract data from websites using libraries like BeautifulSoup or Scrapy.

```
import requests
from bs4 import BeautifulSoup

url = "https://example.com"

response = requests.get(url)
soup = BeautifulSoup(response.text, "html.parser")

print("Page Title:", soup.title.text)

for heading in soup.find_all("h1"):
    print("Heading:", heading.text)
```

Sample Output:

```
Page Title: Example Domain
Heading: Example Domain
```

Note: Install the required libraries with: `pip install requests beautifulsoup4`

2. Hangman: Implement the word-guessing game with visual progress and hints.

```
import random

words = {
    "python": "A programming language",
    "computer": "An electronic machine",
    "school": "A place where students learn"
}

word = random.choice(list(words.keys()))
hint = words[word]
guessed = set()
wrong = 0
max_wrong = 6

stages = [
    "[------]",
    "[0-----]",
    "[0|-----]",
    "[0|/-----]",
    "[0|/\\--]",
    "[0|/\\\\-]",
    "[0|/\\\\/-]",
    "[0|/\\\\/-\\]"
]

print("Hint:", hint)

while wrong < max_wrong:
    display = " ".join(
        letter if letter in guessed else "_" for letter in word
    )
    print("\nWord:", display)
    print("Progress:", stages[wrong])

    if "_" not in display:
        print("You won!")
        break

    guess = input("Guess a letter: ").lower()

    if len(guess) != 1 or not guess.isalpha():
        print("Please enter one letter.")
        continue

    if guess in guessed:
        print("You already guessed that letter.")
        continue

    guessed.add(guess)

    if guess in word:
        print("Correct guess!")
    else:
        wrong += 1
        print("Wrong guess!")
        print("Attempts left:", max_wrong - wrong)

else:
    print("\nProgress:", stages[wrong])
    print("You lost! The word was:", word)
```

Sample Output:

Hint: A programming language

Word: _ _ _ _ _
Progress: [------]

Guess a letter: p
Correct guess!

Word: p _ _ _ _
Progress: [-----]
Guess a letter: y
Correct guess!

Word: p y _ _ _
Progress: [-----]
Guess a letter: t
Correct guess!

Word: p y t _ _
Progress: [-----]
Guess a letter: h
Correct guess!

Word: p y t h _ _
Progress: [-----]
Guess a letter: o
Correct guess!

Word: p y t h o _
Progress: [-----]
Guess a letter: n
Correct guess!
You won!